MA1522 Linear Algebra for Computing

Lesson 4: Inverses, Elementary Matrices and Their Uses

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson4` (e.g. `A1234567Z_lesson4.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 4**.
- Deadline: today, **10:30 AM**.
- If you do not know how to upload your submission, come to ask me after class.

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Basic rules:

- Please (mainly) ask the questions related to the course.

Example 1

Solve the linear system

$$\begin{cases} x_1 + 2x_2 + 3x_3 = 1 \\ 2x_1 + 5x_2 + 3x_3 = 2 \\ x_1 \qquad\quad + 8x_3 = 3 \end{cases}$$

Lesson 2's method — Gauss–Jordan on the augmented matrix:

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 2 & 5 & 3 & 2 \\ 1 & 0 & 8 & 3 \end{array}\right) \xrightarrow[R_3 - R_1]{R_2 - 2R_1} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 1 & -3 & 0 \\ 0 & -2 & 5 & 2 \end{array}\right) \xrightarrow{R_3 + 2R_2} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & -1 & 2 \end{array}\right)$$

$$\xrightarrow{-R_3} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 1 & -2 \end{array}\right) \xrightarrow[R_1 - 3R_3]{R_2 + 3R_3} \left(\begin{array}{ccc|c} 1 & 2 & 0 & 7 \\ 0 & 1 & 0 & -6 \\ 0 & 0 & 1 & -2 \end{array}\right) \xrightarrow{R_1 - 2R_2} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 19 \\ 0 & 1 & 0 & -6 \\ 0 & 0 & 1 & -2 \end{array}\right)$$

All three variable columns are pivot, so the solution is unique:

$$x_1 = 19, \qquad x_2 = -6, \qquad x_3 = -2.$$

Section 2.4 Inverse of Matrices

A square matrix A of order n is **invertible** if there is a matrix B with

$$AB = I_n = BA \qquad \text{(both products, and then } B \text{ is } n \times n \text{ too).}$$

Such a B is unique — if $AB = I = BA$ and $AC = I = CA$ then $B = B(AC) = (BA)C = C$ — so we may write $B = A^{-1}$, **the** inverse of A. A square matrix with no inverse is called **singular**.

Remark

A matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ of order 2 is invertible if and only if $ad - bc \neq 0$, and in that case

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

The generalisation of this formula to order n is called **Cramer's formula**, and we will meet it in Section 2.10.

Section 2.4 Properties of the Inverse, and How to Find It

Let A, B be invertible of order n and $c \neq 0$.

- $(A^{-1})^{-1} = A$;
- $(cA)^{-1} = \frac{1}{c}A^{-1}$;
- $(A^T)^{-1} = (A^{-1})^T$;
- $(AB)^{-1} = B^{-1}A^{-1}$ — the order reverses, exactly as for the transpose;
- $A^{-n} := (A^{-1})^n$.

The algorithm. Finding A^{-1} means solving $AX = I_n$, which by block multiplication is n systems with the *same* coefficient matrix. So run Gauss–Jordan once, on $\left(A \mid I_n\right)$:

$$\left(A \mid I\right) \xrightarrow{\text{Gauss–Jordan}} \left(I \mid A^{-1}\right)$$

If a zero row ever appears in the left block, stop: the RREF of A is not I, and A is singular.

Question 1

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ — the coefficient matrix and the constant vector of Example 1.

1. Find A^{-1}.
2. Hence solve $A\mathbf{x} = \mathbf{b}$.
3. Without any further row reduction, write down $(A^T)^{-1}$ and $(3A)^{-1}$.

Question 1 — solution: part (1)

Run Gauss–Jordan on $(A \mid I_3)$:

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_3 - R_1]{R_2 - 2R_1} \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_3 + 2R_2} \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1 & -5 & 2 & 1 \end{array}\right) \xrightarrow{-R_3} \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right)$$

$$\xrightarrow[R_1 - 3R_3]{R_2 + 3R_3} \left(\begin{array}{ccc|ccc} 1 & 2 & 0 & -14 & 6 & 3 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right) \xrightarrow{R_1 - 2R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right)$$

The left block reached I_3, so A is invertible and

$$A^{-1} = \begin{pmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{pmatrix}.$$

Question 1 — solution: parts (2) and (3)

(2) A is invertible, so the solution exists, is unique, and is $\mathbf{x} = A^{-1}\mathbf{b}$:

$$\mathbf{x} = \begin{pmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} -40 + 32 + 27 \\ 13 - 10 - 9 \\ 5 - 4 - 3 \end{pmatrix} = \begin{pmatrix} 19 \\ -6 \\ -2 \end{pmatrix}.$$

Which is exactly what Example 1 got by row reduction. ✓ The inverse did not give a *better* answer — it gave the *same* answer with all the work on A already done.

(3) Straight from the properties, no work at all:

$$(A^T)^{-1} = (A^{-1})^T = \begin{pmatrix} -40 & 13 & 5 \\ 16 & -5 & -2 \\ 9 & -3 & -1 \end{pmatrix}, \qquad (3A)^{-1} = \tfrac{1}{3}A^{-1} = \tfrac{1}{3}\begin{pmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{pmatrix}.$$

Note $(3A)^{-1} \neq 3A^{-1}$: scaling A up scales the inverse *down*. □

Section 2.5 Elementary Matrices

An **elementary matrix** E is what you get by doing one elementary row operation to I_n. The point of them: $A \xrightarrow{r} EA$ — doing r to A = premultiplying A by E.
Here is an example.

$$EA = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \mathbf{r}_1 \\ \mathbf{r}_2 \\ \mathbf{r}_3 \end{pmatrix} = \begin{pmatrix} 1\mathbf{r}_1 + 0\mathbf{r}_2 + 0\mathbf{r}_3 \\ 3\mathbf{r}_1 + 1\mathbf{r}_2 + 0\mathbf{r}_3 \\ 0\mathbf{r}_1 + 0\mathbf{r}_2 + 1\mathbf{r}_3 \end{pmatrix} = \begin{pmatrix} \mathbf{r}_1 \\ 3\mathbf{r}_1 + \mathbf{r}_2 \\ \mathbf{r}_3 \end{pmatrix} \quad \text{— exactly } R_2 + 3R_1.$$

- Every elementary matrix is invertible, and E^{-1} is the elementary matrix of the **reverse** operation — here $E^{-1} = \left(\begin{smallmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right)$, the matrix of $R_2 - 3R_1$:

$$R_i + cR_j \;\leftrightarrow\; R_i - cR_j, \qquad R_i \leftrightarrow R_j \text{ is its own reverse}, \qquad cR_i \;\leftrightarrow\; \tfrac{1}{c}R_i.$$

- Hence if $A \xrightarrow{r_1} \cdots \xrightarrow{r_k} B$ with elementary matrices $E_1, \ldots, E_k$, then

$$B = E_k \cdots E_1 A \qquad \text{and} \qquad A = E_1^{-1} E_2^{-1} \cdots E_k^{-1} B.$$

Both orders reverse. Read the second one as: undo r_k first.

Question 2

A 3×3 matrix A and a vector $\mathbf{b}$ are not given to you. All you are told is what three row operations do to them:

$$\begin{pmatrix} ? & ? & ? \\ ? & ? & ? \\ ? & ? & ? \end{pmatrix} = A \xrightarrow{R_2 - 3R_1} \xrightarrow{R_3 + 5R_1} \xrightarrow{R_3 - R_2} U = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 2 & -1 \\ 0 & 0 & 0 \end{pmatrix},$$

$$\begin{pmatrix} ? \\ ? \\ ? \end{pmatrix} = \mathbf{b} \xrightarrow{R_2 - 3R_1} \xrightarrow{R_3 + 5R_1} \xrightarrow{R_3 - R_2} \mathbf{c} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

1. Find a matrix M with $A = MU$.
2. Hence solve $A\mathbf{x} = \mathbf{b}$, still without working out what A and $\mathbf{b}$ are.
3. Now work out A and $\mathbf{b}$, and check your answer to (2).

Question 2 — solution: part (1)

Let E_1, E_2, E_3 be the elementary matrices of $R_2 - 3R_1$, $R_3 + 5R_1$, $R_3 - R_2$. Then $E_3E_2E_1A = U$, so

$$A = E_1^{-1}E_2^{-1}E_3^{-1}U = MU, \qquad M := E_1^{-1}E_2^{-1}E_3^{-1}.$$

Each E_i^{-1} is the reverse operation, so

$$M = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -5 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ -5 & 1 & 1 \end{pmatrix}.$$

Question 2 — solution: parts (2) and (3)

(2) The same operations act on $\mathbf{b}$, so $E_3E_2E_1\mathbf{b} = \mathbf{c}$, that is $\mathbf{b} = M\mathbf{c}$. As M is invertible (a product of elementary matrices),

$$A\mathbf{x} = \mathbf{b} \iff MU\mathbf{x} = M\mathbf{c} \iff U\mathbf{x} = \mathbf{c}.$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 0 & 2 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right) \longrightarrow \left(\begin{array}{ccc|c} 1 & 0 & 5/2 & -1/2 \\ 0 & 1 & -1/2 & 1/2 \\ 0 & 0 & 0 & 0 \end{array}\right) \implies \mathbf{x} = \begin{pmatrix} -1/2 \\ 1/2 \\ 0 \end{pmatrix} + s\begin{pmatrix} -5 \\ 1 \\ 2 \end{pmatrix}, \quad s \in \mathbb{R}.$$

(3) $A = MU$ and $\mathbf{b} = M\mathbf{c}$:

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ -5 & 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 & 2 \\ 0 & 2 & -1 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 2 \\ 3 & 5 & 5 \\ -5 & -3 & -11 \end{pmatrix}, \qquad \mathbf{b} = M\mathbf{c} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}.$$

Check: $A\begin{pmatrix} -1/2 \\ 1/2 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ ✓ and $A\begin{pmatrix} -5 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ ✓. □

Section 2.6 Equivalent Statements for Invertibility

Theorem

For a square matrix A of order n, the following are **equivalent** — each one implies all the others.

1. A is invertible;
2. A^T is invertible;
3. $BA = I$ for some B (a **left inverse**);
4. $AB = I$ for some B (a **right inverse**);
5. the RREF of A is I_n;
6. A is a product of elementary matrices;
7. $A\mathbf{x} = \mathbf{0}$ has only $\mathbf{x} = \mathbf{0}$;
8. $A\mathbf{x} = \mathbf{b}$ is uniquely solvable, for every $\mathbf{b}$.

Challenge

Let A and B be square matrices of the same size. Show that if $AB = I$, then $BA = I$. (You may use the equivalence above.)

Example 2

True or false? Let A and B be square matrices of the same size. If AB is invertible, then A and B are both invertible.

True. Write $C = (AB)^{-1}$, so that $C(AB) = I = (AB)C$. Regroup each product — matrix multiplication is associative, which is all we need:

$$I = C(AB) = (CA)B \qquad \text{and} \qquad I = (AB)C = A(BC).$$

The first says B has a **left inverse**, namely CA; the second says A has a **right inverse**, namely BC. By statements (3) and (4) of the theorem, both B and A are invertible.

Question 3

Let A be a 3×3 matrix, and suppose you are told only that each of the three systems

$$A\mathbf{x} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \qquad A\mathbf{x} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \qquad A\mathbf{x} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

is consistent. Which of the following must be true? Justify your answer.

1. $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution.
2. There is some $\mathbf{b}$ for which $A\mathbf{x} = \mathbf{b}$ has infinitely many solutions.
3. Every REF of A has three nonzero rows.
4. There is not enough information to decide (1), (2) or (3).

Question 3 — solution

Let $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$ be solutions of the three systems and put $X = \begin{pmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 \end{pmatrix}$. By block multiplication (Lesson 3), the three statements bundle into one:

$$AX = \begin{pmatrix} A\mathbf{u}_1 & A\mathbf{u}_2 & A\mathbf{u}_3 \end{pmatrix} = C, \qquad C = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix} \xrightarrow[\text{Gauss–Jordan}]{} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

So the RREF of C is I_3, and by (5)⇒(1) of the theorem C is invertible. Then

$$A\big(XC^{-1}\big) = (AX)C^{-1} = CC^{-1} = I_3,$$

so A has a *right* inverse — and by (4)⇒(1), A is invertible. Now read the list:

(1) **False** by (7). (2) **False** by (8) — every **b** gives exactly one solution. (3) **True**: the RREF of A is I_3 by (5); a REF with a zero row would keep that zero row through Gauss–Jordan, and the RREF is unique. (4) **False** — three consistent systems were plenty.

Only **(3)**. Note what did the work: three *separate* systems became one *matrix* equation. □

Challenge

A square matrix A is **nilpotent** if $A^k = 0$ for some $k \geq 1$; for instance $A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$ has $A^3 = 0$.

Show that for such an A, the matrix $I - A$ is invertible, and find $(I - A)^{-1}$.

Guess from the geometric series $\frac{1}{1-x} = 1 + x + x^2 + \cdots$, which here terminates. Put $S = I + A + A^2 + \cdots + A^{k-1}$. The product telescopes:

$$(I - A)S = (I + A + \cdots + A^{k-1}) - (A + A^2 + \cdots + A^k) = I - A^k = I,$$

and the same computation gives $S(I - A) = I$. So $I - A$ is invertible with $(I - A)^{-1} = S$.

In fact one of the two products would have sufficed: by (4)⇒(1) of Section 2.6, a right inverse of a square matrix is automatically its inverse.

For the example above, $(I - A)^{-1} = I + A + A^2 = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$.

Next lesson

- **Sections 2.7–2.9 LU Factorization and Determinants**

Please watch the video clips for these sections beforehand.

If you do not know how to upload your submission, come to ask me after class.